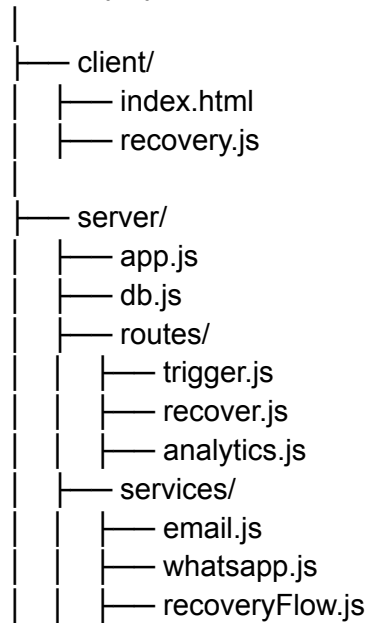


📁 PROJECT STRUCTURE

recovery-system/



.env

package.json

📦 package.json

```
{
  "name": "recovery-system",
  "version": "1.0.0",
  "main": "server/app.js",
  "scripts": {
    "start": "node server/app.js"
  },
  "dependencies": {
    "cors": "^2.8.5",
    "dotenv": "^16.0.0",
    "express": "^4.18.2",
    "pg": "^8.11.0",
    "@sendgrid/mail": "^7.7.0",
    "twilio": "^4.0.0"
  }
}
```

🔑 .env

SENDGRID_API_KEY=your_sendgrid_key

EMAIL_FROM=your@email.com

TWILIO_SID=your_sid

TWILIO_AUTH=your_auth

TWILIO_WHATSAPP_NUMBER=whatsapp:+14155238886

🧠 DATABASE (RUN THIS)

CREATE TABLE sessions (
 id SERIAL PRIMARY KEY,

```
    session_id VARCHAR(50),
    cart JSON,
    trigger_type VARCHAR(20),
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

```
CREATE TABLE recoveries (
    id SERIAL PRIMARY KEY,
    email VARCHAR(100),
    cart JSON,
    status VARCHAR(20),
    sent_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);
```

 BACKEND

 server/db.js

```
const { Pool } = require("pg");
```

```
const pool = new Pool({
  user: "postgres",
  host: "localhost",
  database: "recovery_db",
  password: "password",
  port: 5432,
});
```

```
module.exports = pool;
```

 server/app.js

```
const express = require("express");
```

```
const cors = require("cors");
```

```
require("dotenv").config();
```

```
const triggerRoute = require("./routes/trigger");
```

```
const recoverRoute = require("./routes/recover");
```

```
const analyticsRoute = require("./routes/analytics");
```

```
const app = express();
```

```
app.use(cors());
```

```
app.use(express.json());
```

```
app.use("/api/trigger", triggerRoute);
```

```
app.use("/api/recover", recoverRoute);
```

```
app.use("/api/analytics", analyticsRoute);
```

```
app.listen(3000, () => {
```

```
  console.log("Server running on port 3000");
```

```
});
```

 server/routes/trigger.js

```

const express = require("express");
const router = express.Router();
const pool = require("../db");

router.post("/", async (req, res) => {
  const { sessionId, cart, triggerType } = req.body;

  try {
    await pool.query(
      "INSERT INTO sessions (session_id, cart, trigger_type) VALUES ($1, $2, $3)",
      [sessionId, cart, triggerType]
    );

    res.json({ status: "tracked" });
  } catch (err) {
    res.status(500).json({ error: "DB error" });
  }
});

```

```

module.exports = router;
server/routes/recover.js
const express = require("express");
const router = express.Router();
const pool = require("../db");
const { startRecoveryFlow } = require("../services/recoveryFlow");

```

```

router.post("/", async (req, res) => {
  const { email, phone, cart } = req.body;

  try {
    await pool.query(
      "INSERT INTO recoveries (email, cart, status) VALUES ($1, $2, $3)",
      [email, cart, "started"]
    );

    startRecoveryFlow(email, phone, cart);

    res.json({ status: "flow_started" });
  } catch (err) {
    res.status(500).json({ error: "Recovery failed" });
  }
});

```

```

module.exports = router;
server/routes/analytics.js
const express = require("express");
const router = express.Router();
const pool = require("../db");

```

```

router.get("/", async (req, res) => {
  const sessions = await pool.query("SELECT COUNT(*) FROM sessions");
  const recoveries = await pool.query("SELECT COUNT(*) FROM recoveries");

  res.json({
    total_sessions: sessions.rows[0].count,
    total_recoveries: recoveries.rows[0].count
  });
});

```

```

module.exports = router;

```

⚙️ SERVICES

📁 server/services/email.js

```

const sgMail = require("@sendgrid/mail");

```

```

sgMail.setApiKey(process.env.SENDGRID_API_KEY);

```

```

const sendRecoveryEmail = async (to, cart) => {
  const items = cart.map(i => `<li>${i.name} - ₹${i.price}</li>`).join("");

```

```

  const msg = {
    to,
    from: process.env.EMAIL_FROM,
    subject: "Complete your purchase 🛒",
    html: `
      <h2>You left items in your cart</h2>
      <ul>${items}</ul>
      <p>Come back now and get a discount!</p>
    `
  };

```

```

  await sgMail.send(msg);
};

```

```

module.exports = { sendRecoveryEmail };

```

📁 server/services/whatsapp.js

```

const twilio = require("twilio");

```

```

const client = new twilio(
  process.env.TWILIO_SID,
  process.env.TWILIO_AUTH
);

```

```

const sendWhatsApp = async (to, cart) => {
  const items = cart.map(i => `${i.name} - ₹${i.price}`).join(", ");

```

```

  await client.messages.create({

```

```

    from: process.env.TWILIO_WHATSAPP_NUMBER,
    to: `whatsapp:${to}`,
    body: `🛒 You left items: ${items}. Complete your order now!`
  });
};

```

```

module.exports = { sendWhatsApp };
📁 server/services/recoveryFlow.js
const { sendRecoveryEmail } = require("../email");
const { sendWhatsApp } = require("../whatsapp");

const startRecoveryFlow = (email, phone, cart) => {

  setTimeout(() => {
    sendRecoveryEmail(email, cart);
  }, 10 * 60 * 1000);

  setTimeout(() => {
    if (phone) sendWhatsApp(phone, cart);
  }, 30 * 60 * 1000);

  setTimeout(() => {
    sendRecoveryEmail(email, cart);
  }, 2 * 60 * 60 * 1000);

  setTimeout(() => {
    sendRecoveryEmail(email, cart);
  }, 24 * 60 * 60 * 1000);
};

```

```

module.exports = { startRecoveryFlow };

```

🌐 FRONTEND

📁 client/index.html

HTML

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Recovery System</title>
```

```
</head>
```

```
<body>
```

```
<h1>Demo Store</h1>
```

```
<button class="add-to-cart" data-id="1" data-name="Product A" data-price="500">
```

```
Add to Cart
```

```
</button>
```

```
<div id="recovery-popup" style="display:none;">
```

```

<h2>Wait! Get Discount</h2>
<input type="email" id="user-email" placeholder="Email" />
<input type="text" id="user-phone" placeholder="WhatsApp" />
<button onclick="submitData()">Get Offer</button>
</div>

```

```

<script src="recovery.js"></script>

```

```

<script>
function submitData() {
  const email = document.getElementById("user-email").value;
  const phone = document.getElementById("user-phone").value;
  const cart = JSON.parse(localStorage.getItem("cart")) || [];

  fetch("http://localhost:3000/api/recover", {
    method: "POST",
    headers: {"Content-Type": "application/json"},
    body: JSON.stringify({ email, phone, cart })
  });

  alert("Recovery started!");
}
</script>

```

```

</body>

```

```

</html>

```

```

📁 client/recovery.js

```

```

class RecoveryEngine {
  constructor() {
    this.cart = JSON.parse(localStorage.getItem("cart")) || [];
    this.sessionId = "sess_" + Math.random().toString(36).substr(2);
    this.idleTime = 0;
    this.popupShown = false;

    this.init();
  }

  init() {
    this.trackCart();
    this.detectExit();
    this.idleTimer();
  }

  trackCart() {
    document.querySelectorAll(".add-to-cart").forEach(btn => {
      btn.addEventListener("click", e => {
        const item = {
          id: e.target.dataset.id,

```

```

        name: e.target.dataset.name,
        price: e.target.dataset.price
    };

    this.cart.push(item);
    localStorage.setItem("cart", JSON.stringify(this.cart));
    });
    });
}

detectExit() {
    document.addEventListener("mouseleave", e => {
        if (e.clientY <= 0 && !this.popupShown) {
            this.trigger("exit");
        }
    });
}

idleTimer() {
    setInterval(() => {
        this.idleTime++;
        if (this.idleTime > 15 && !this.popupShown) {
            this.trigger("idle");
        }
    }, 1000);

    document.onmousemove = document.onkeypress = () => {
        this.idleTime = 0;
    };
}

async trigger(type) {
    this.popupShown = true;

    await fetch("http://localhost:3000/api/trigger", {
        method: "POST",
        headers: {"Content-Type": "application/json"},
        body: JSON.stringify({
            sessionId: this.sessionId,
            cart: this.cart,
            triggerType: type
        })
    });

    document.getElementById("recovery-popup").style.display = "block";
}
}

```

```
new RecoveryEngine();
```

 RUN SYSTEM

```
npm install
```

```
npm start
```

Open:  client/index.html